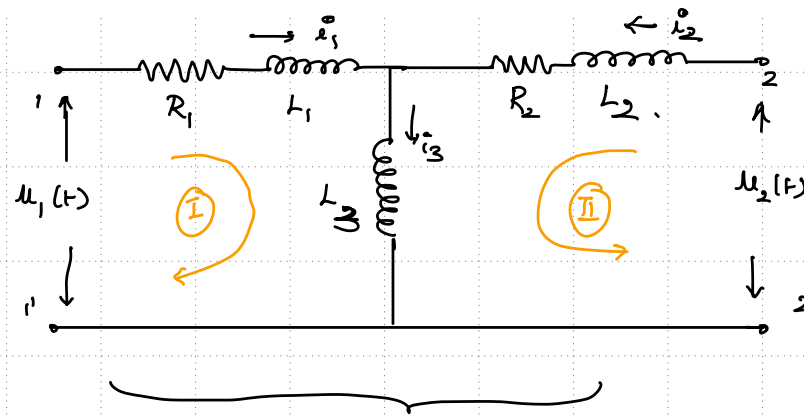

EE1060: Differential Equations and Transform Techniques

Lecture 35: Systems of Differential Equations

Topics :

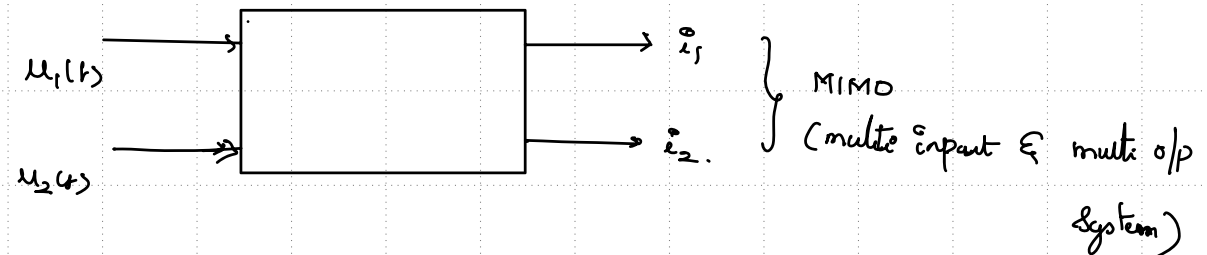
1. Introduction to Systems of Differential Equations
 2. Homogeneous system of Differential Equations
-

Introduction to Systems of Differential Equations

Example:-

independent variable: time (t)

ckt:-

inputs:- $u_1(t)$ & $u_2(t)$ output: $i_1(t)$ & $i_2(t)$ $y_1 = i_1, y_2 = i_2$ KVL to Loop 1:-

$$u_1(t) = R_1 i_1(t) + L_1 \frac{di_1}{dt} + L_3 \frac{di_3}{dt} = R_1 i_1(t) + (L_1 + L_3) \frac{di_1}{dt} + L_3 \frac{di_2}{dt}$$

$$(L_1 + L_3) \frac{di_1}{dt} + L_3 \frac{di_2}{dt} = u_1(t) - R_1 i_1(t) \rightarrow \textcircled{1}$$

Loop 2:-

$$L_3 \frac{di_1}{dt} + (L_2 + L_3) \frac{di_2}{dt} = u_2(t) - R_2 i_2(t) \rightarrow \textcircled{2}$$

$$\begin{bmatrix} L_1 + L_3 & L_3 \\ L_3 & L_2 + L_3 \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -R_1 & 0 \\ 0 & -R_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$$

$\vec{y} = \mathbf{I} \vec{x}$
 Rewritten as
 $\dot{\vec{x}} = \mathbf{A} \vec{x} + \mathbf{B} \vec{u}$

Introduction to Systems of Differential Equations

Example:-

$$a_n(t) \frac{d^n x}{dt^n} + \dots + a_1(t) x'(t) + a_0(t) x(t) = u(t) \rightarrow \textcircled{1}$$

def:- o/p to be
 $x^{(n-1)}(t) +$
 $x'(t)$

def:-

$$\left. \begin{array}{ll} x_1(t) = \dot{x}(t) & \Rightarrow \dot{x}_1(t) = x_2(t) \\ x_2(t) = \dot{x}_1(t) = \ddot{x}(t) & \Rightarrow \dot{x}_2(t) = x_3(t) \\ \vdots & \vdots \\ x_{n-1}(t) = \dot{x}_{n-2}(t) = x^{(n-1)}(t) & \Rightarrow \dot{x}_{n-1}(t) = x_n(t) \\ x_n(t) = \dot{x}_{n-1}(t) = x^{(n)}(t) & \Rightarrow \dot{x}_n(t) = x_{n+1}(t) \end{array} \right\} \rightarrow \textcircled{2}$$

$\vec{y} = \dot{C} \vec{x}$

① can be written as:

$$a_n(t) x_n(t) + \dots + a_1(t) x_1(t) + a_0(t) x(t) = u(t)$$

$$x_n(t) = \frac{1}{a_n(t)} \left[u(t) - \sum_{i=0}^{n-1} a_i(t) x_i(t) \right] \rightarrow \textcircled{3}$$

def:- $\vec{x} = \begin{bmatrix} x(t) \\ x_1(t) \\ \vdots \\ x_{n-1}(t) \end{bmatrix}$ = Vector of n functions

$$\begin{bmatrix} \dot{x} \\ \dot{x}_1 \\ \vdots \\ \dot{x}_{n-1} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\frac{a_0}{a_n} & \dots & -\frac{a_{n-1}}{a_n} \end{bmatrix} \begin{bmatrix} x \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix} + \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \frac{1}{a_n} \end{bmatrix} [u(t)] \Rightarrow \dot{\vec{x}} = A(t) \vec{x} + B(t) \vec{u}$$

functions of t

Solution of a system of Differential Equations - Trajectory

In this course:- $\begin{cases} \dot{\vec{x}} = A\vec{x} + B\vec{u} \rightarrow (1) \\ \vec{y} = C\vec{x} \rightarrow (2) \end{cases}$ where $A, B \& C$ are matrices of functions with independent variable 't'.
 $\vec{x} \& \vec{u}$ vectors of fn of 't'.

To begin with: $A, B \& C$ are Constant matrices

$\vec{x}_s$ is some Region in $\mathbb{R}^n$
Solution of (1) on Interval I:- (or) for some 't' in $\mathbb{R}$ (✓).

A vector of functions $\vec{x}_s$ is a solution to (1) on I if all entries of $\vec{x}_s$ are differentiable on I and satisfy (1).

Example:-

$\vec{x} = \begin{bmatrix} \sin t \\ \cos t \end{bmatrix}$ is a solution of some 2nd order ODE
 represents a circle in $\mathbb{R}^2$
 in general case: solution represents a trajectory.

Linear Independence and Homogeneous System of Differential Equations

Two vectors of functions $\vec{x}(t)$ and $\vec{z}(t)$ are linearly dependent if $\exists \alpha \neq 0$ such that

$$\vec{x}(t) = \alpha \vec{z}(t).$$

For a collection $\{\vec{x}_1(t), \dots, \vec{x}_n(t)\}$ are linearly dependent if $\exists \alpha_1, \dots, \alpha_n$ at least one of which is non-zero such that

$$\sum_{i=1}^n \alpha_i \vec{x}_i(t) = \vec{0} \quad \text{over entire } I.$$

$\uparrow$
 zero vector.